

Hanjun Kim



Ph.D. Candidate @ RPM Robotics lab, being advised by Prof. Ayoung Kim

My research primarily focuses on Autonomous Driving, SLAM, Sensor Fusion, and Deep Learning. Currently, I'm researching gravity-aware radar-visual-inertial SLAM for quadruped robot in adverse conditions.

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EDUCATION

- | | |
|---|-------------------|
| Seoul National University , Republic of Korea | 03/2026 – |
| • Ph.D. Candidate in Department of Mechanical Engineering | |
| Seoul National University , Republic of Korea | 03/2024 – 02/2026 |
| • M.S. student in Department of Future Automotive Mobility | |
| • Related Courses: [Machine Learning for Visual Understanding], [Future Automotive Mobility Engineering], [Autonomous Driving], [Electric Vehicle Engineering] | |
| • Cumulative GPA: 4.36/4.5 | |
| Technical University of Munich , Germany | 03/2023 – 08/2023 |
| • Exchange student in Department of Mechanical Engineering | |
| • Related Courses: [Introduction to Deep learning], [Autonomous Driving] | |
| Seoul National University , Republic of Korea | 03/2018 – 02/2024 |
| • Bachelor in Department of Mechanical Engineering | |
| • Related Courses: [Global Engineering Internship], [Theory and Lab IoT, AI, Big Data], [Automotive engineering], [Mechanical System Modeling and Control] | |
| • Cumulative Major GPA: 3.94/4.5 | |

RESEARCH EXPERIENCE

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|---|-------------------|
| Robust Perception and Mobile Robotics Lab | 03/2024 – Present |
| • Legged Robot with camera-radar-IMU fusion system for robust perception | |
| • Heterogeneous radar place recognition for cross-modal localization | |
| • Heterogeneous radar dataset in complex urban environment for multi-session radar SLAM | |
| • National R&D Project for Smart Construct Technology to develop autonomous driving UGV system with LiDAR-based SLAM, path planning and path tracking module. | |
| Artificial Intelligence Semiconductor Engineering Lab | 09/2022 – 02/2023 |
| • Autonomous driving project internship | |
| • Analyzed and reviewed research and papers about Stereo vision, Semi-Global Matching | |
| • Dynamic 3D mapping using zed2i stereo camera with ROS1 | |
| Innovative Design and Integrated Manufacturing Lab | 09/2022 – 12/2023 |
| • Autonomous collaborative robot system based on voice command | |
| • Construction of a system to control Doosan's robot arm using voice commands API and ROS1 | |
| ICRA 2024 Late-Breaking Result Poster Presentation | 05/2024 |
| • Vision-based collaborative robot automation and voice control system using mobile robot arms | |

EXTERNAL ACTIVITIES

Sequor Robotics, Robot software and hardware solutions startup **April 2023 – August 2023**

- Industry-academic cooperation project.
- Pedestrian trajectory prediction algorithm development.

SUM, Self-driving startup Internship Field practice **January 2023 – February 2023**

- Practice on positioning process using Kalman filter based on sensor data.
- Distant object tracking algorithm using Lidar, Radar and Vision.
- CARLA simulation open scenario production.
- PID controller design practice using Simulink to implement lateral and longitudinal control.

Hyundai America Technical Center, Inc. Headquarters Global Internship **June 2022 – July 2022**

- Benchmarking, development projects in several departments including AD department.
- Lab tour and experience various departments.

Visual Camp, eye tracking technology startup Internship **May 2022 – June 2022**

- App planner/developer.
- Business proposal to enter the global market.

Research Assistant of Hyundai, SNU's Industrial Cooperation Group **March 2022 – June 2022**

- Produce a report that summarizes the latest technology trends.
- Class assistant work and class-related administrative work.

Sigma Intelligence, a robot club in Electrical Engineering **March 2019 – December 2020**

- Made health trainer robot using Gyro sensor, ultrasonic sensor, and Python.

PUBLICATIONS

- H. Kim et al., "Hercules: Heterogeneous radar dataset in complex urban environment for multi-session radar slam," in Proc. of the IEEE International Conference on Robotics and Automation, 2025.
- H. Kim et al., "SHerLoc: Synchronized Heterogeneous Radar Place Recognition for Cross-Modal Localization." IEEE Robot. and Automat. Lett., 2025.

Other careers

- 2024.03` Excellent Research Award, Dept. of System Semiconductor Engineering for AI, SNU
- 2024.01` TOEIC Speaking test AL(Advanced Low: score 160)
- 2023.12` Outstanding BS Thesis Award, Dept. of Mechanical Engineering, SNU
- 2023.11` Kaggle ML4ME Object detection challenge 1st place
- 2022.12` Seoul National University x Hyundai Doosan Infracore Mechatronics Contest Grand Prize
- 2020.02` Microsoft Office Specialist Master
- 2019.05` Seoul National University GECE The Action Leadership Award
- 2017.12` The 7th U.S. Embassy Seoul UCC Contest Semi-finalist
- SNU's startup club SNUSV's 30th generation
- Artificial Intelligence Camp Assistant Instructor(likelion x newsac)
- MiraeAsset Park Hyunjoo Foundation 28th Scholarship Student